

SASWAT K. NAYAK

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EXPERIENCE

Data Scientist(Full Time) Innovaccer Analytics(Digital Tech Unicorn)

July 2020-July 2022

- Convinced the clinical leaders to shift the entire knowledge base DB(PostgreSQL) containing mappings to ICD, SNOMED codes to a graph database(Neo4j) leading to 80% reduction in query time.
- Led a team of interns to extract patient data from progress notes, EHR data, achieving a 90% reduction in time and a 50% decrease in costs by implementing NLP solutions(NLTK, Spacy) and using auto-encoders for anomaly detection.
- Optimized deep learning models using NVIDIA TensorRT to accelerate inference speed by up to 5x for deployment in production environments.
- Orchestrated the automatic deployment and scaling of a containerized task queue subsystem responsible for batch processing across dozen of worker nodes- Docker, EC2 for healthcare provider's recommendation systems.
- Designing and Implementing Machine Learning Pipelines: Developing and maintaining scalable machine learning pipelines to ensure seamless integration and deployment of models.

Projects [Streamlit Visualization](#)

- Execute an end-to-end data engineering [project](#) on real-time stock market data using Amazon Web Services (AWS), Apache Kafka, Glue, Athena, and SQL.
- Utilized **LangChain** and **Llama2** to develop a [chatbot](#) fine-tuned on my experience. Enhanced its performance by 70% through the implementation of RAG, embeddings, vector database, and optimized document chunking techniques.
- Fine-tuned Mistral-7B on GPT-4 generated rationales, enabling reasoning in Mistral-7B by knowledge distillation. Fine-tuned Mistral-7B on the call classification dataset by using AWQ quantization, LoRA and SFT.
- Built an emergency SOS Voice Agent with LlamaIndex Agent RAG and Bolna Voice Agent.
- Scraped Nordstrom using Selenium, rotating proxies to prevent website from scrape-blocking and multi-threading for improved scraping speeds- [Github](#).

EDUCATION

Master of Science, Artificial Intelligence in 3D Photonics

September 2022-November 2024

RELEVANT COURSES

Software Engineering, Image Processing, Advanced Machine Learning, Statistics, Information Retrieval, Databases, Probability and Stochastic Processes, Regression and Time Series Analysis, Machine Learning in Materials, Algorithms and Data Structures, Linear Algebra, Graph Databases, Generative AI.

SKILLS

Javascript, Python, CI/CD, Docker, Gitlab,R, C++, SQL, Tableau, Triton, TensorRT , Spark, GPU, FastAPI, Angular, React, Java, Agile, Redis, GCL, Mage, Azure, Caboodle, A/B testing, Excel, Power BI, Hugging Face, Operating Systems, Computer Graphics, Node.js, RDkit Data, AWS Batch, NLP, GANs, RNN, Gen AI, Tensorflow, PyTorch, GANs, Airflow, Statistics, LLM, Pandas, Numpy, Spacy, Large Language Models(BERT, GPT-2, T5, RoBERTa), Google TTS, GBM, XGBoost, AWS Bedrock, Flink, Langchain, Transformers, MLops, Streamlit, Knowledge Graphs, Large Distributed Systems, JAX, vLLM

GRA: Generative AI in 3D Photonics, Semiconductor Design

September 2022 - Present 2023 — *Madison, WI*

- Introduced novel spatial descriptors for 3D lattices that captured more information than the traditional symmetry based descriptors and outperformed them in machine learning tasks by 50%(F1 Score).
- Made a mixture of **generative AI models** capable of generating 3D lattices with a wide band gap.

Machine Learning Intern, Ushur

Nov 2020 - Dec 2020

- Designed and implemented an advanced algorithm combining Google Tesseract OCR with rule-based techniques to accurately extract customer data from insurance claim documents; securing victory in an internal hackathon.